EvolveR: Self-Evolving LLM Agents through an Experience-Driven Lifecycle

Rong Wu, Xiaoman Wang et al. · arXiv 2510.16079v3 · Independent Reel interpretation

HarnessRSI 65 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Kept: principle library + policy weights
Advice + triples + linked source trajectories; historical usage and success counts per entry

1 Online interaction
Think → search experience / knowledge → answer
Store successful and failed training trajectories

2 Offline self-distillation
Freeze policy; same model writes guiding or cautionary advice + structured knowledge triples

3 Add novel entries, merge duplicates
Embedding prefilter (0.85) + semantic yes/no; merge equivalent trajectories under one principle

4 Keep by historical utility
score = (successes + 1) / (uses + 2)
Periodic sweep: prune principles below 0.3

5 Update policy weights, repeat
GRPO: exact-answer correctness + format reward
Next tasks retrieve top-3 retained principles

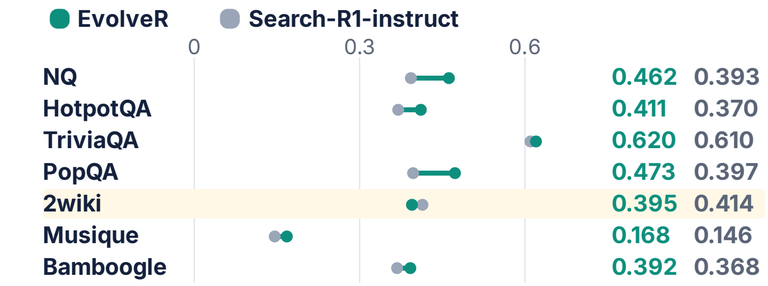
Not memory-only: context AND weights evolve

Experience source: NQ + HotpotQA training splits. GPT-4o supplies cold-start trajectories with gold answers; later principle writing is self-distilled.

2 Higher average, uneven gains

3B: 0.382
vs 0.325 · Search-R1-instruct

7B: 0.417
vs 0.385 · Search-R1-instruct



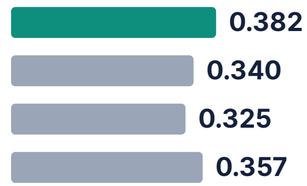
7B EM · Table 1. Strongest baseline in each column. Largest gains: NQ / PopQA; 2wiki loses. Seven-benchmark averages; runs, SD / CI unreported.

4 Where the gain is absent / thin

- **3B:** TriviaQA **0.584 vs 0.587**; PopQA **0.434 vs 0.435** (Search-R1-base). Bamboogle only 0.328 vs Search-o1 0.320.
- **7B:** 2wiki **0.395 vs 0.414**; TriviaQA 0.620 vs 0.610. Small margins have no uncertainty estimates.
- **Memory growth:** validation EM 0.410 at 45,083 principles, then 0.387 at 49,934 (Table 7). Policy also changes; not a memory-size-only test.

3 Controls and protocol boundaries

3B average EM · Tables 3–4



Retrieval-off: SAME experience-trained policy, disabled only at evaluation. Distinct from no-distillation RL.

Same memory mechanism, different writer · Table 2

Qwen scale	self	teacher
0.5B	0.150	**0.220**
1.5B	0.270	**0.290**
3B	**0.382**	0.370

GPT-4o-mini teacher. At 3B, teacher still wins 2wiki: 0.437 vs 0.381. No universal self-distillation advantage.

- **Budget:** 3B lifecycle 39.4 h on 8 A100s. Baseline costs / tuning budgets, separate curation / teacher costs unreported.
- **Memory parity:** same curated store not reported for Table 1 baselines. SFT shares successful trajectories: 0.357 vs GRPO 0.382; equal compute not established.
- **Splits / leakage:** OOD declared eval-only; no overlap audit / explicit eval-writeback rule. Exact eval splits and checkpoint/default-selection split unspecified.

Takeaway

Retained principles help beyond the trained weights: disabling retrieval lowers the same policy's score. But the headline combines memory and RL, self-distillation loses at smaller scales, and neither equal experience budgets nor leakage boundaries are fully established.

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Reports: cross-task tests, retrieval ablation, teacher control, runtime (judgment addendum).
Not reported: seed variance / CI, matched budgets, overlap audit.